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Department of Examinations

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Acknowledgments and Abstract

I would like to express my heartfelt gratitude to my lecturers for their invaluable guidance, encouragement, and support throughout the development of this project.

I am deeply thankful to **Dr. Tharaka Sankalapa** and **Mr. Sanjaya Suraweera** for their expert insights and continuous mentorship, which helped me navigate complex technical challenges and refine my work to meet industry standards. Their feedback and encouragement played a pivotal role in shaping GLORO into a complete, feature-rich food delivery application.

Abstract

GLORO is a next-generation food delivery application designed to transform the way users experience online ordering. With a sleek, intuitive interface and carefully curated features, GLORO empowers customers to discover restaurants, explore food options, customize orders, and seamlessly complete payments through a secure gateway.

The app extends beyond typical delivery systems by offering flexible navigation styles, order tracking with location services, and personalized recommendations through local data storage. Sellers can efficiently manage products, track orders, and analyze sales trends, while administrators oversee platform activities to ensure a smooth ecosystem.

Built with a robust technology stack including Android, SQLite, Firestore, and a Java backend with Hibernate . GLORO bridges the gap between customer convenience and restaurant efficiency. The result is a dynamic, responsive application that not only simplifies food delivery but elevates the overall user experience. This report delves into the design, implementation, and impact of GLORO, showcasing how thoughtful development can lead to a practical, high-impact solution for the food delivery industry.

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1. Introduction

The Gloro food delivery app is designed to revolutionize the way users order food by providing a **fast, efficient, and user-friendly platform**. This document outlines the requirements, objectives, and scope of the Gloro app, ensuring a clear understanding of its purpose and functionality. The app aims to address the growing demand for convenient and reliable food delivery services while offering a seamless experience for both customers and restaurant partners.

1.1 Monitoring

The motivation behind developing Gloro stems from the increasing reliance on technology to simplify everyday tasks. In today's fast-paced world, people are looking for ways to save time and effort, especially when it comes to ordering food. Traditional methods of food delivery often involve long wait times, limited options, and a lack of transparency in the ordering process.

Gloro aims to bridge these gaps by offering a **modern, intuitive, and feature-rich platform** that caters to the needs of both customers and restaurants. By leveraging advanced technologies such as real-time order tracking, secure payment gateways, and personalized user experiences, Gloro seeks to provide a solution that is not only convenient but also enjoyable to use.

Additionally, the app is designed to empower small and medium-sized restaurants by giving them a platform to reach a wider audience, manage orders efficiently, and grow their businesses. Gloro's mission is to create a **win-win situation** where customers get their favorite meals delivered quickly, and restaurants benefit from increased visibility and sales.

1.2 Problem Statement

Despite the popularity of food delivery apps, several challenges persist in the current market:

1. **Limited Customization**: Many existing apps do not allow users to customize their orders (e.g., adding toppings or special instructions), leading to a less satisfying experience.
2. **Inefficient Cart Management**: Users often face difficulties when ordering from multiple restaurants or managing their cart items. The lack of advanced cart features, such as organizing items by shop or adjusting quantities, can lead to confusion and errors.
3. **Poor User Experience**: Many apps have cluttered interfaces, slow performance, and limited personalization options, making them difficult to navigate and use.
4. **Limited Support for Restaurants**: Small and medium-sized restaurants often struggle to compete with larger chains due to limited visibility and inefficient order management systems.

Gloro addresses these challenges by providing a **comprehensive solution** that prioritizes user convenience, customization, and transparency. By offering features like advanced cart management, order tracking, and a user-friendly interface, Gloro aims to set a new standard for

food delivery apps.

2.Background

Food delivery apps have become a part of our daily lives. Whether it's a busy professional grabbing lunch between meetings or a family ordering dinner on a lazy Sunday, these apps have made it easier than ever to enjoy restaurant-quality meals at home. But let's be honest—while food delivery apps are convenient, they're not perfect. That's where **Gloro** comes in. Gloro isn't just another food delivery app; it's a **game-changer** designed to solve the frustrations users face with existing platforms.

2.1 The Evolution of Food Delivery Apps

Food delivery has come a long way from the days of phone orders and limited restaurant options. With the rise of smartphones and on-demand services, the industry has transformed into a fast-paced, tech-driven ecosystem. Today, users expect more than just a menu and a checkout button—they want **speed**, **transparency**, and **personalization**.

However, many existing platforms focus on scaling their services without addressing the core needs of their users. This has created a gap in the market for a more **user-centric solution**, which is where Gloro comes in.

2.2 Gloro's Unique Approach

Gloro is designed to address these challenges by offering a **modern, intuitive, and feature-rich platform** that prioritizes the needs of both customers and restaurants. Here's what sets Gloro apart:

1. **Customizable Orders**:
   Gloro lets users tailor their orders to their exact preferences. Whether it's adding extra toppings, specifying cooking instructions, or choosing portion sizes, Gloro gives users the freedom to order food their way.
2. **Smart Cart Management**:
   The app's **smart cart system** organizes items by restaurant, making it easy to adjust quantities, remove items, or save them for later. It's designed to simplify the ordering process, even when ordering from multiple places.
3. **Personalized Experience**:
   Gloro is all about **you**. The app offers customizable themes, navigation options, and a **wishlist** feature, allowing users to save their favorite dishes for future orders. It's designed to feel personal and engaging.
4. **Empowering Restaurants**:
   Gloro isn't just for customers it's for restaurants too. The **seller app** provides tools for menu

management, order tracking, and performance analysis, helping restaurants grow their businesses and serve their customers better.

3.Specification

The purpose of this section is to provide a clear picture of the **Gloro** food delivery app, outlining its **functional requirements**, **non-functional requirements**, and **constraints**. The specification describes what the system is required to do, while the design (covered in the next section) will explain how the system meets these requirements.

3.1Functional Requirements

Functional requirements define the specific features and functionalities of the Gloro app. These are the core capabilities that the app must deliver to meet user needs.

1. **User Registration and Login**:
    - Users can create an account or log in using their email credentials.
    - Secure authentication ensures user data is protected.
2. **Menu Display and Navigation**:
    - Users can browse food categories and view detailed information about each item, including name, description, ingredients, price, and images.
    - Filters and sorting options allow users to find food items based on preferences (e.g., price range, categories).
3. **Order Placement and Cart Management**:
    - Users can add food items to their cart, adjust quantities, and remove items.
    - The cart organizes items by restaurant, making it easy to manage orders from multiple shops.
    - Users can proceed to checkout when ready to place their order.
4. **Payment Integration**:
    - The app supports secure payment methods, including debit cards and cash payments.
5. **Order Tracking (Current Capability)**:
    - Users can track their orders by viewing the **shop location** and **delivery address**.
    - While real-time tracking is not currently available, this feature is planned for future updates.
6. **Push Notifications**:
    - Users receive notifications about placed order.
7. **User Profile and Order History**:
    - Users can update their profile information, including name and profile picture.

- Order history allows users to view past orders and reorder their favorite items.

8. **Wishlist**:
    - Users can save food items to a wishlist for future orders.
9. **Customizable App Experience**:
    - Users can choose between **bottom navigation** or **drawer navigation** for the app layout.
10. **Admin Panel (For Restaurants)**:
    - Restaurants can manage their menus, update product details and update order status.
    - The admin panel provides tools for managing foods and monitoring sales.

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system, such as performance, security, and usability.

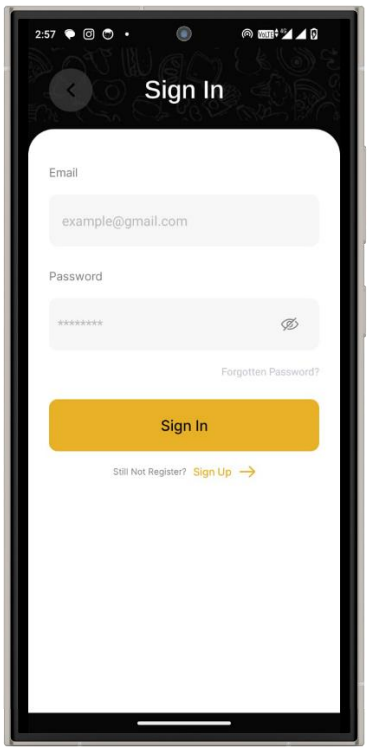
1. **Performance**:
    - The app should load quickly and respond to user interactions within 3 seconds.
2. **Scalability**:
    - The app should be designed to accommodate future updates, such as real-time order tracking and additional features.
    - It should support an increasing number of users and restaurants without requiring significant changes to the architecture.
3. **Security**:
    - User data, including login credentials and payment information, must be encrypted.
4. **Compatibility**:
    - The app should be compatible with Android devices running Android 6.0 (Marshmallow) and above.
5. **Usability**:
    - The app should have an intuitive and user-friendly interface.
    - Navigation should be simple and consistent across all screens.

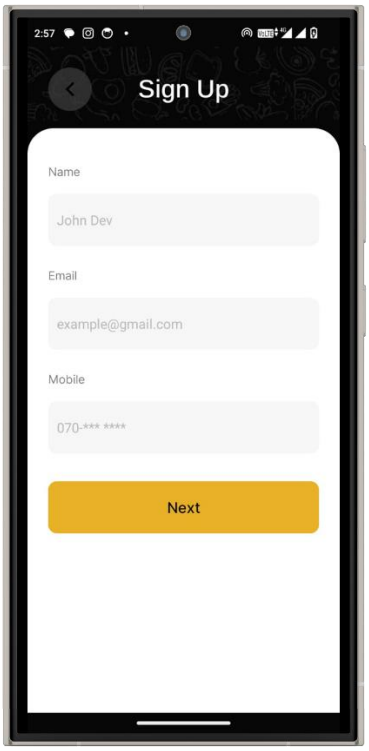
4.Design

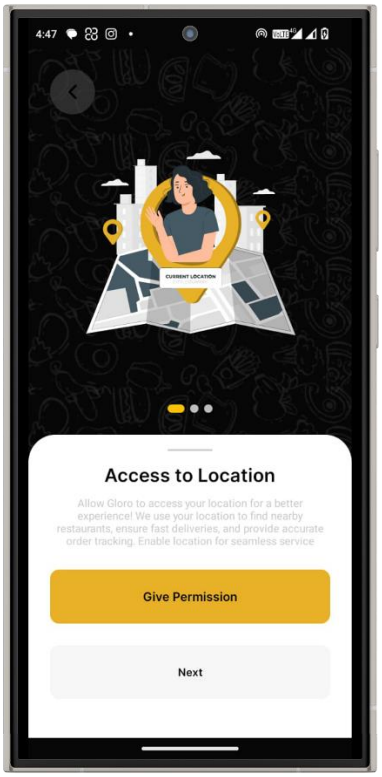
The design of the Gloro food delivery app focuses on creating a **visually appealing**, **intuitive**, and **user-friendly** experience. This section describes the **UI/UX design**, **key screens**, and **visual elements** of the app, supported by images that showcase the app's layout and functionality.

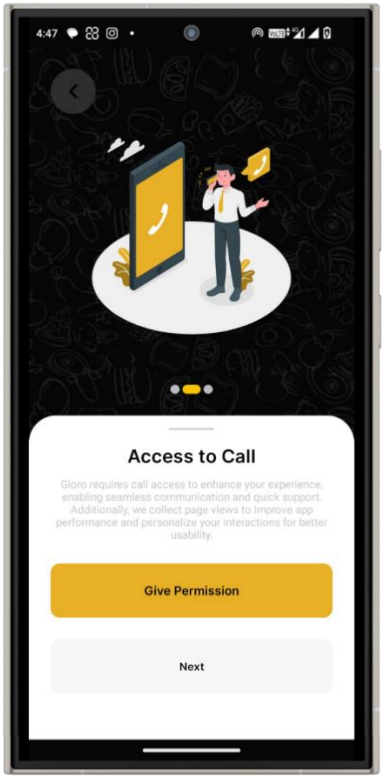
4.1User App Interface

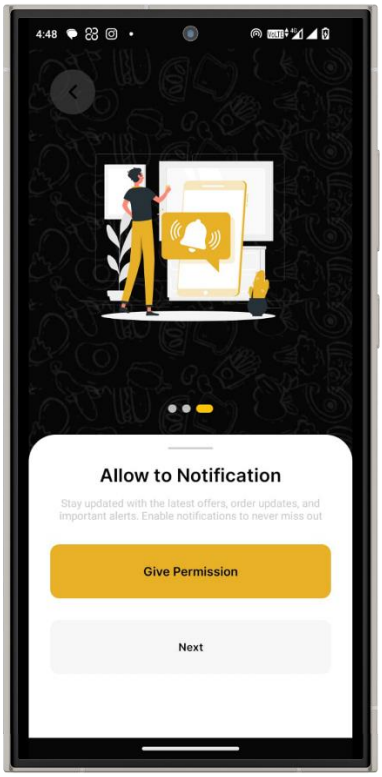


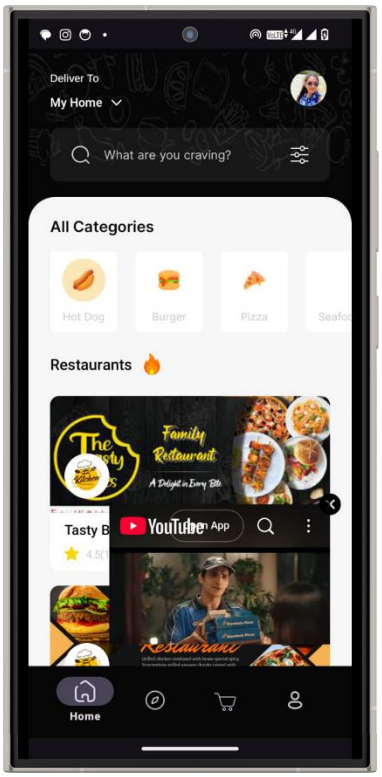


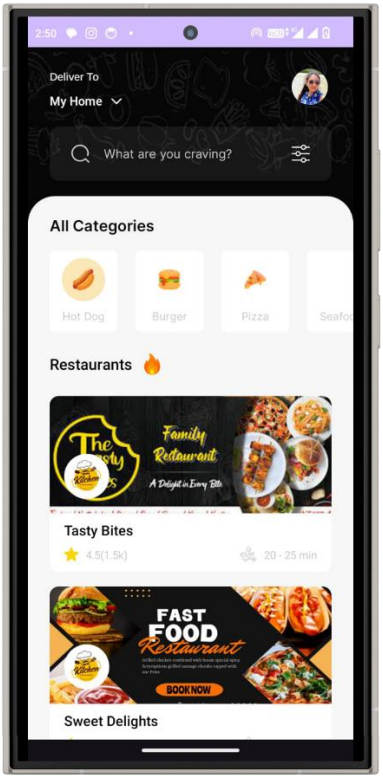


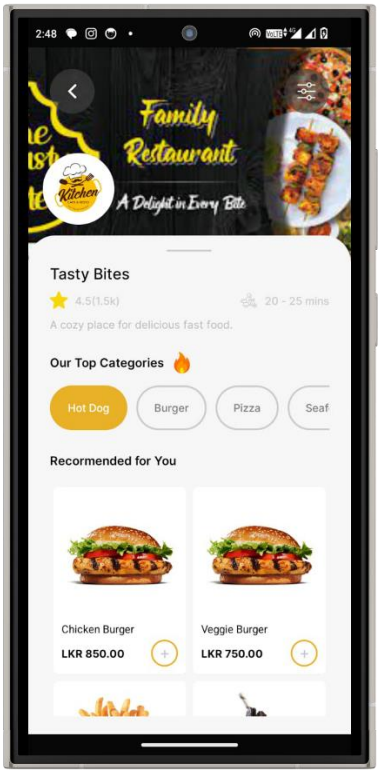

Access to Location
Allow Gloro to access your location for a better experience! We use your location to find nearby restaurants, ensure fast deliveries, and provide accurate order tracking. Enable location for seamless service
Give Permission
Next

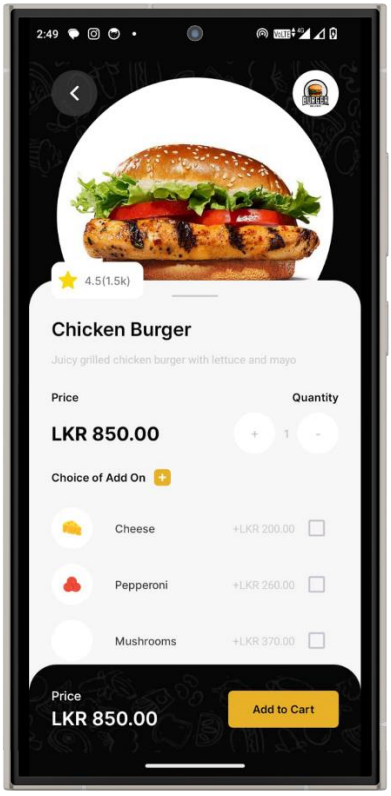

Access to Call
Gloro requires call access to enhance your experience, enabling seamless communication and quick support. Additionally, we collect page views to improve app performance and personalize your interactions for better usability.
Give Permission
Next

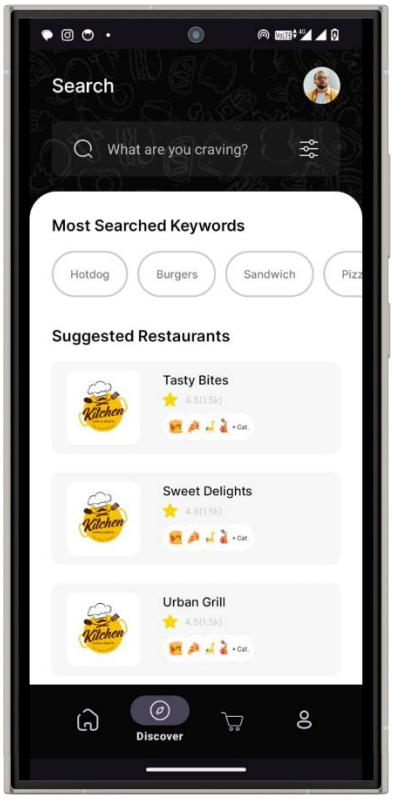

Allow to Notification
Stay updated with the latest offers, order updates, and important alerts. Enable notifications to never miss out
Give Permission
Next

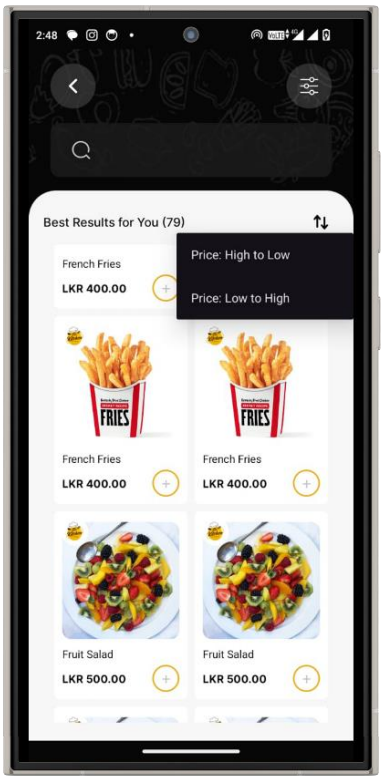

Deliver To
My Home
What are you craving?
All Categories
Hot Dog
Burger
Pizza
Restaurants
Family Restaurant
A Delight in Every Bite
Tasty B
YouTube
Restaurant
Home

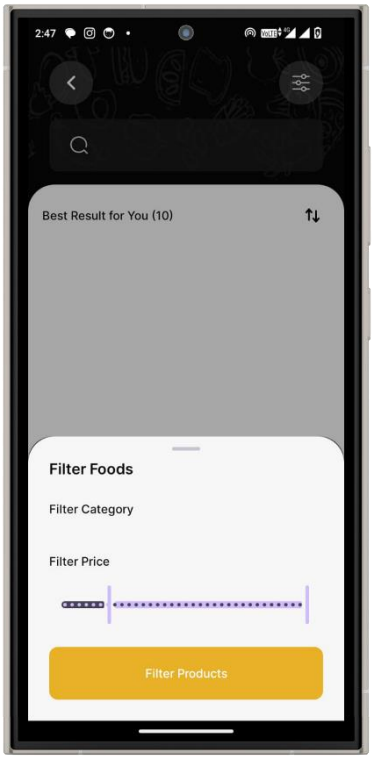

Deliver To
My Home
What are you craving?
All Categories
Hot Dog
Burger
Pizza
Restaurants
Family Restaurant
A Delight in Every Bite
Tasty Bites
20 - 25 min
FAST FOOD
Restaurant
BOOK NOW
Sweet Delights

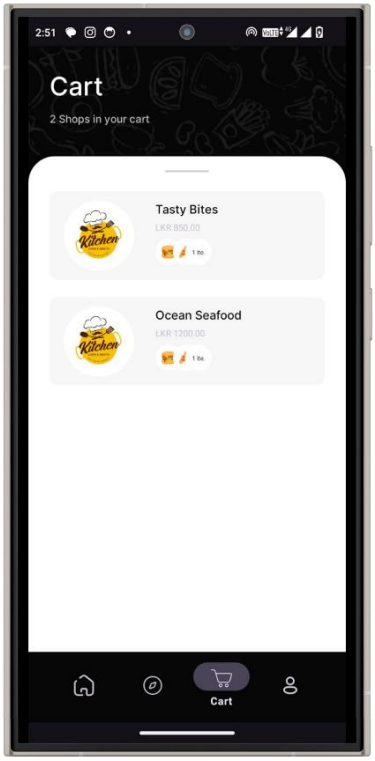

Family Restaurant
A Delight in Every Bite
Tasty Bites
20 - 25 mins
A cozy place for delicious fast food.
Our Top Categories
Hot Dog
Burger
Pizza
Recommended for You
Chicken Burger
LKR 850.00
Veggie Burger
LKR 750.00

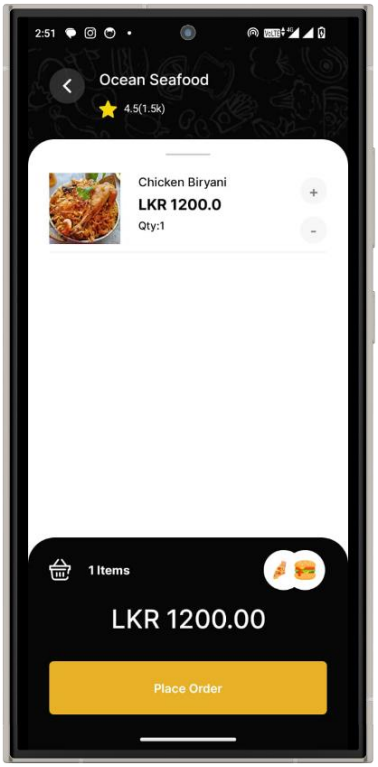

4.5(1.5k)
Chicken Burger
Juicy grilled chicken burger with lettuce and mayo
Price
Quantity
LKR 850.00
Choice of Add On
Cheese
+LKR 200.00
Pepperoni
+LKR 260.00
Mushrooms
+LKR 370.00
Price
LKR 850.00
Add to Cart

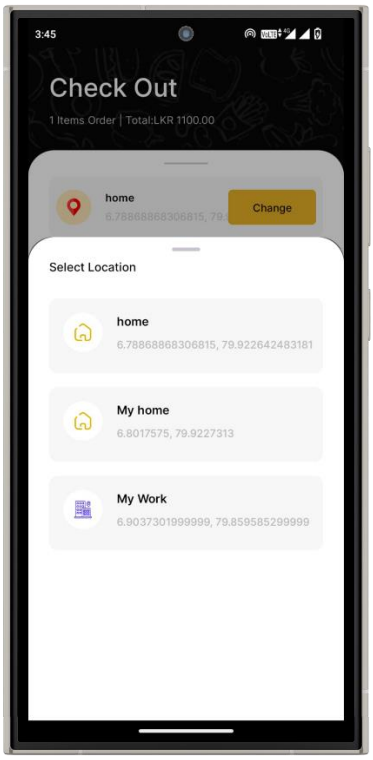

Search
What are you craving?
Most Searched Keywords
Hotdog
Burgers
Sandwich
Suggested Restaurants
Tasty Bites
Sweet Delights
Urban Grill
Discover

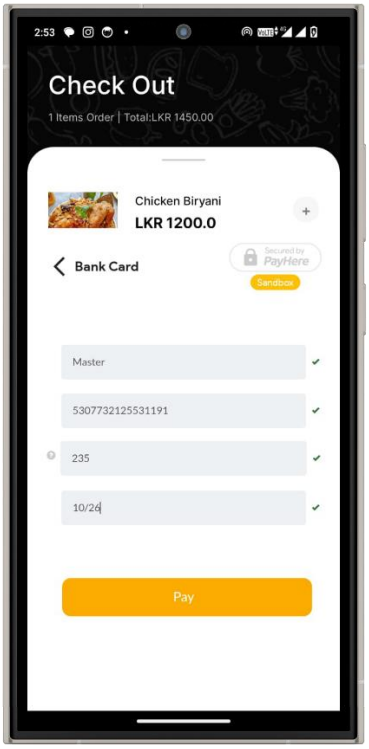

Best Results for You (79)
Price: High to Low
Price: Low to High
French Fries
LKR 400.00
Fruit Salad
LKR 500.00

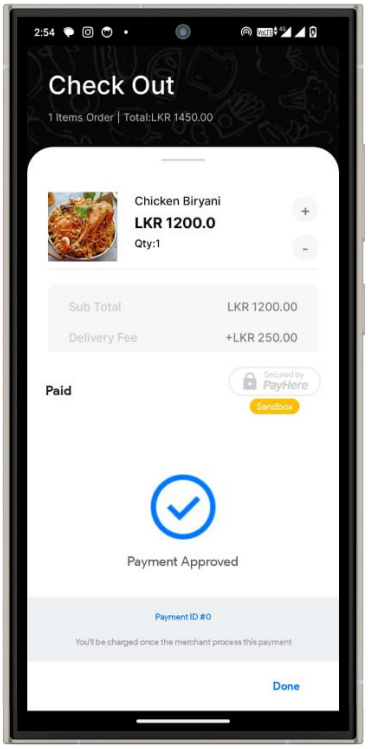

Best Result for You (10)
Filter Foods
Filter Category
Filter Price
Filter Products

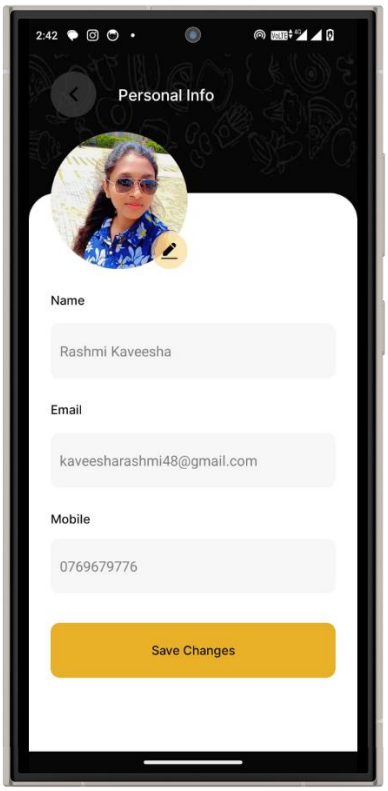

Cart
2 Shops in your cart
Tasty Bites
Ocean Seafood
Cart

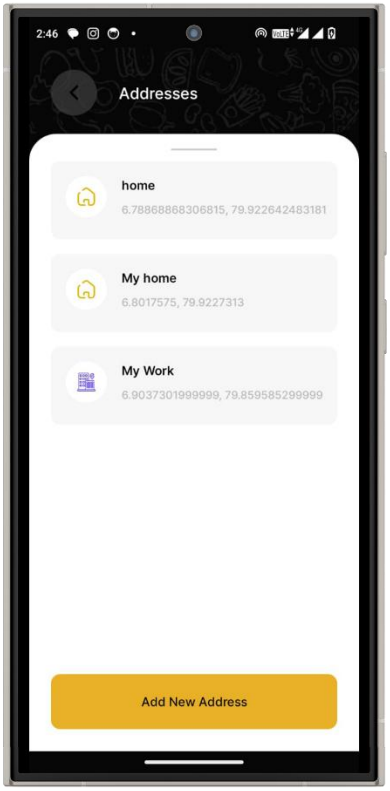

Ocean Seafood
4.5(1.5k)
Chicken Biryani
LKR 1200.0
Qty:1
1 Items
LKR 1200.00
Place Order

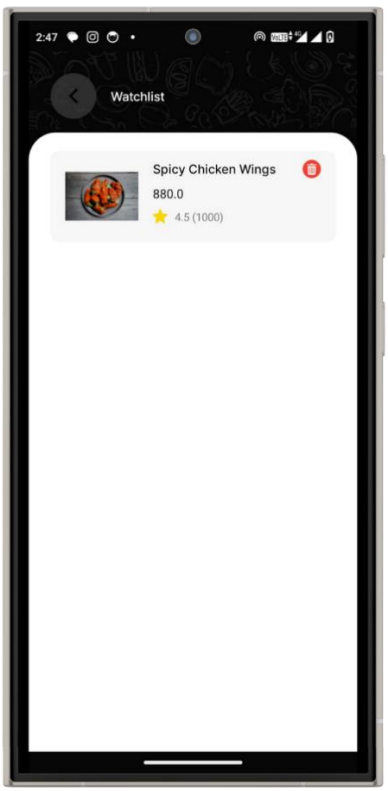

3:45
Check Out
1 Items Order | Total:LKR 1100.00
home
Change
Select Location
home
6.78868868306815, 79.922642483181
My home
6.8017575, 79.9227313
My Work
6.9037301999999, 79.859585299999

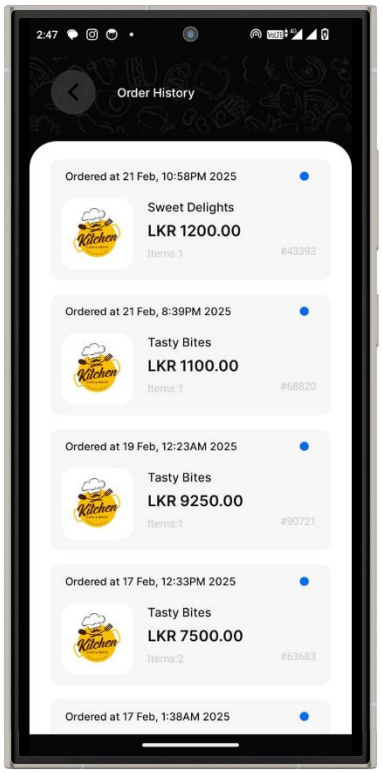

2:53
Check Out
1 Items Order | Total:LKR 1450.00
Chicken Biryani
LKR 1200.0
Bank Card
Secured by PayHere
Sandbox
Master
5307732125531191
235
10/26
Pay

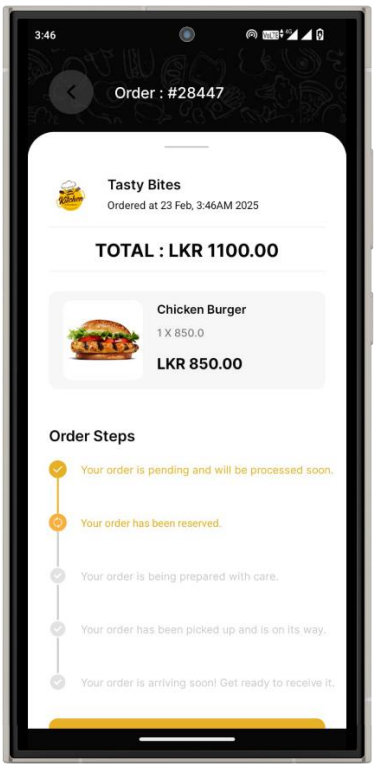

2:54
Check Out
1 Items Order | Total:LKR 1450.00
Chicken Biryani
LKR 1200.0
Qty:1
Sub Total
LKR 1200.00
Delivery Fee
+LKR 250.00
Paid
Secured by PayHere
Sandbox
Payment Approved
Payment ID #0
You'll be charged once the merchant process this payment
Done


2:42
Personal Info
Name
Rashmi Kaveesha
Email
kaveesharashmi48@gmail.com
Mobile
0769679776
Save Changes

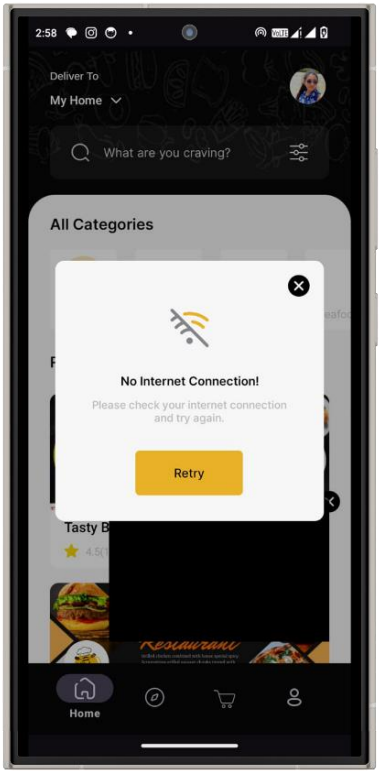

2:46
Addresses
home
6.78868868306815, 79.922642483181
My home
6.8017575, 79.9227313
My Work
6.9037301999999, 79.859585299999
Add New Address

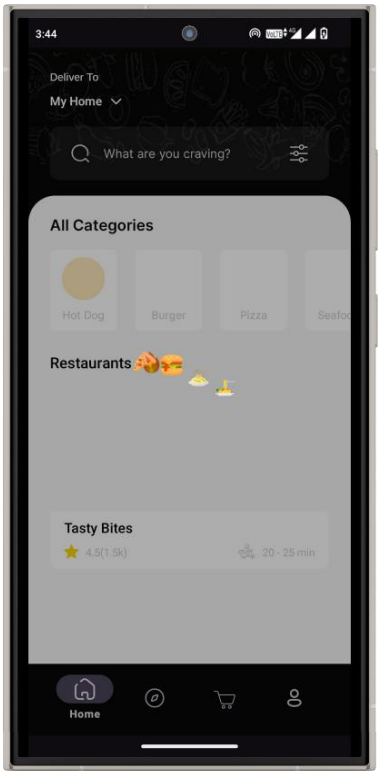

2:47
Watchlist
Spicy Chicken Wings
880.0
4.5 (1000)

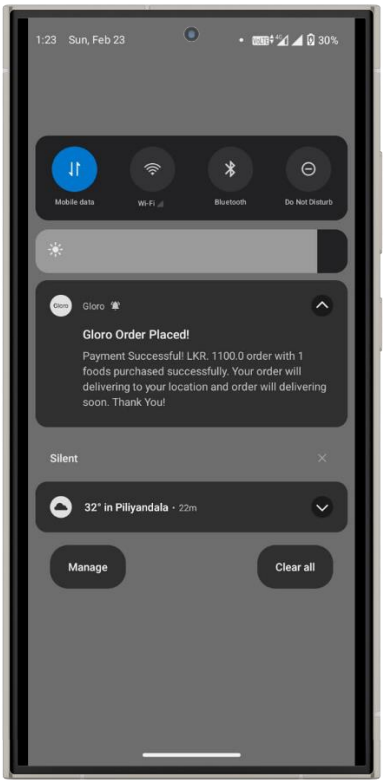

Order History
Ordered at 21 Feb, 10:58PM 2025
Sweet Delights
LKR 1200.00
Ordered at 21 Feb, 8:39PM 2025
Tasty Bites
LKR 1100.00
Ordered at 19 Feb, 12:23AM 2025
Tasty Bites
LKR 9250.00
Ordered at 17 Feb, 12:33PM 2025
Tasty Bites
LKR 7500.00
Ordered at 17 Feb, 1:38AM 2025


Order : #28447
Tasty Bites
Ordered at 23 Feb, 3:46AM 2025
TOTAL : LKR 1100.00
Chicken Burger
1 X 850.0
LKR 850.00
Order Steps
Your order is pending and will be processed soon.
Your order has been reserved.
Your order is being prepared with care.
Your order has been picked up and is on its way.
Your order is arriving soon! Get ready to receive it.


Tasty Bites
0752345678
Order Accepted
Taken
Done
24km
Distance of the Restuarunt

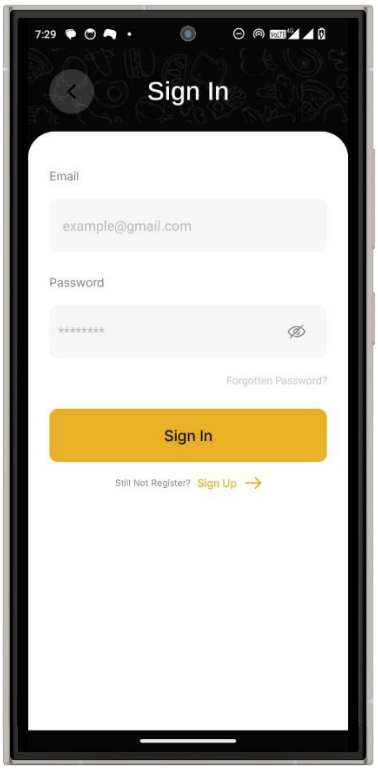

Deliver To
My Home
What are you craving?
All Categories
No Internet Connection!
Please check your internet connection and try again.
Retry
Home

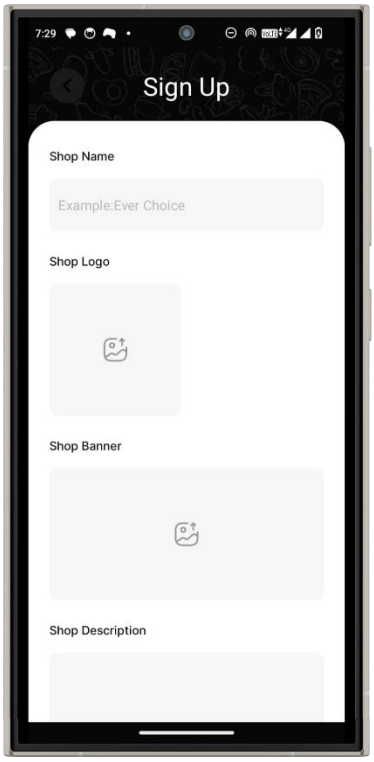

Deliver To
My Home
What are you craving?
All Categories
Hot Dog
Burger
Pizza
Restaurants
Tasty Bites
Home

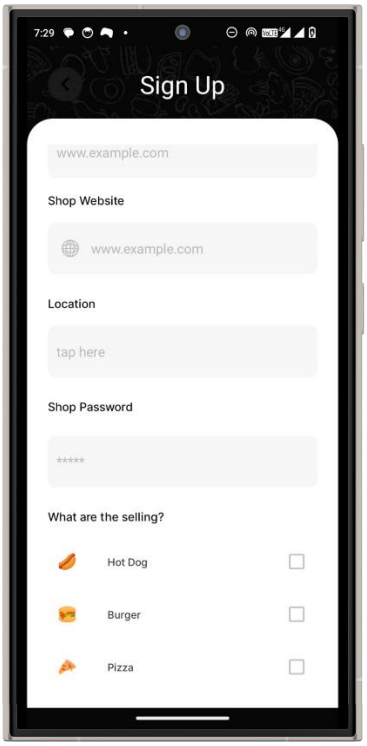

Sun, Feb 23
Mobile data
Wi-Fi
Bluetooth
Do Not Disturb
Gloro
Gloro Order Placed!
Payment Successful! LKR. 1100.0 order with 1 foods purchased successfully. Your order will delivering to your location and order will delivering soon. Thank You!
Silent
32° in Piliyandala
Manage
Clear all

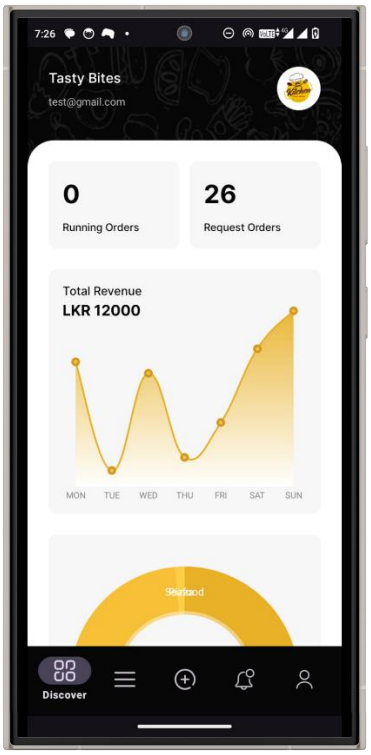
4.2 Seller App Interface

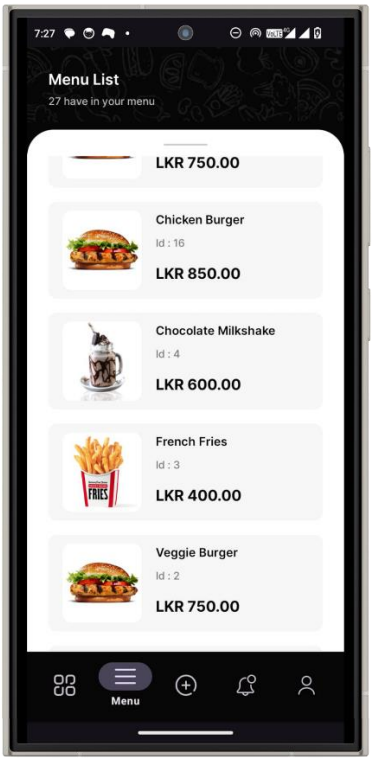


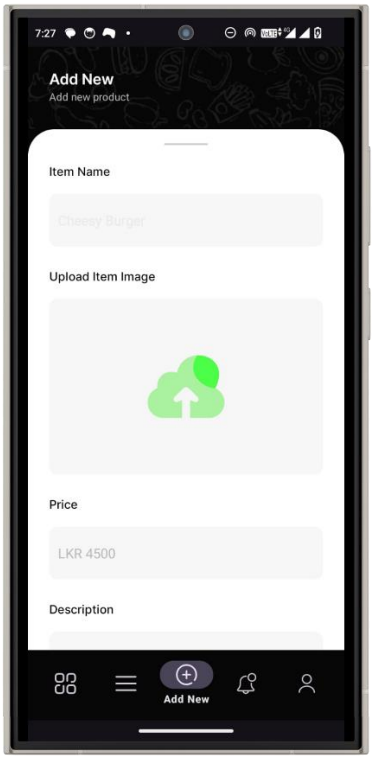


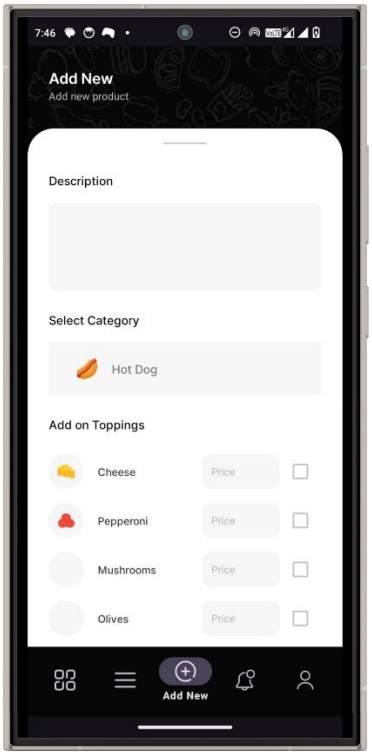


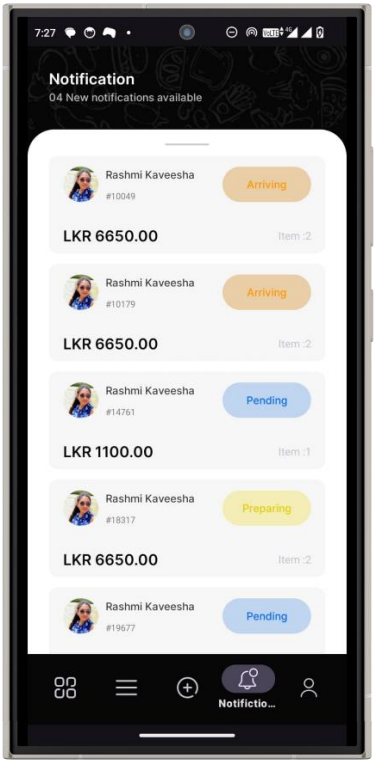


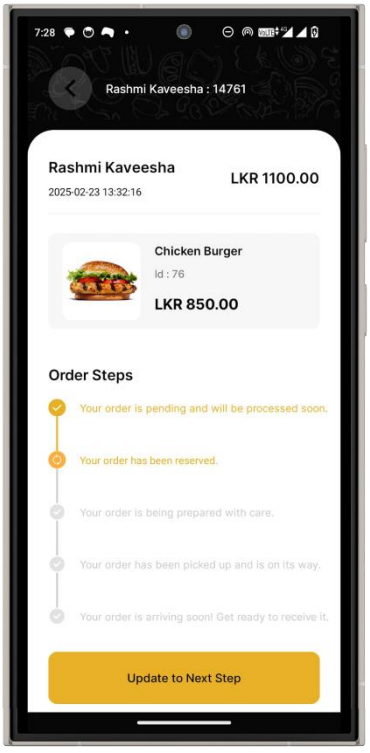


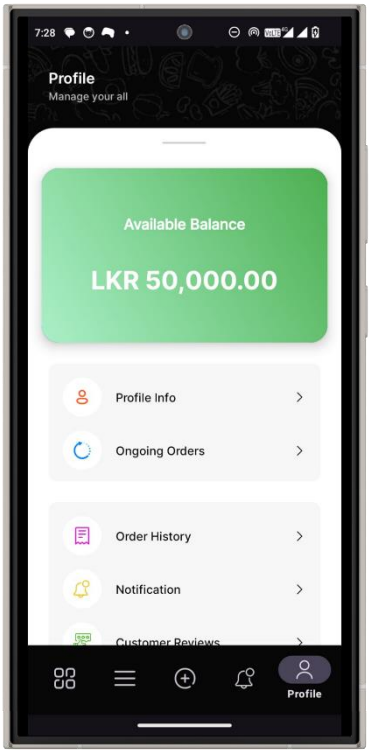



Add New
Add new product
Item Name
Cheesy Burger
Upload Item Image
Price
LKR 4500
Description
Add New

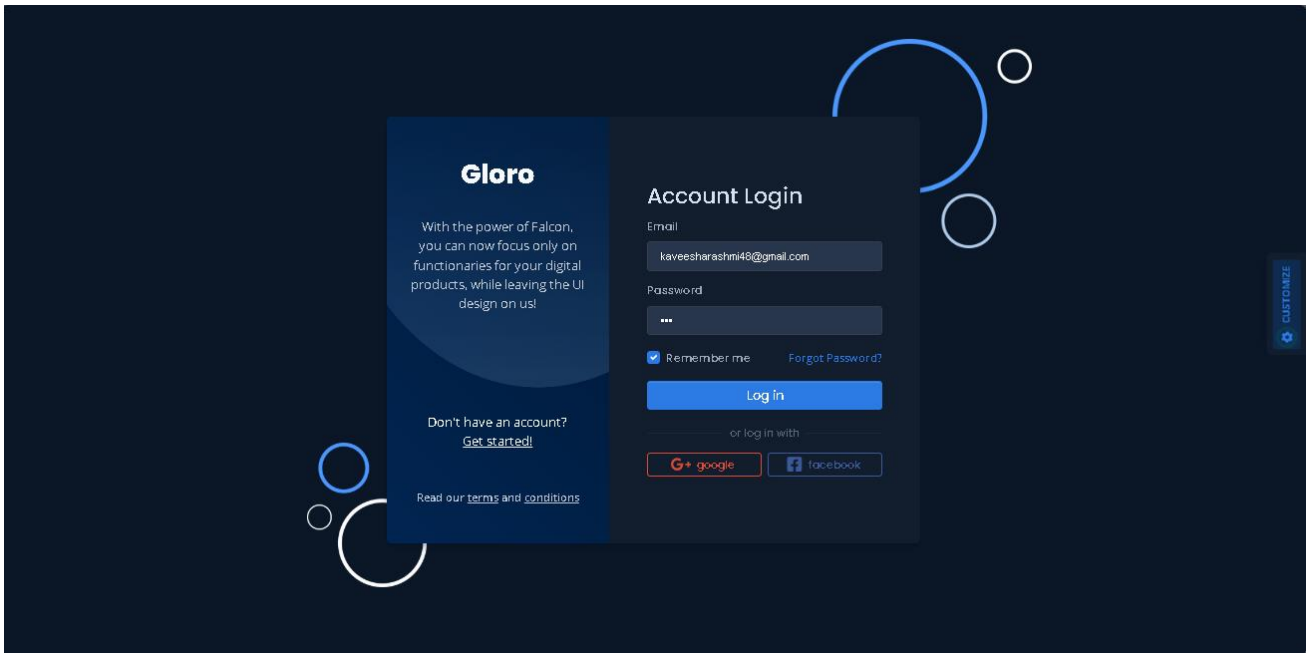

Add New
Add new product
Description
Select Category
Hot Dog
Add on Toppings
Cheese
Pepperoni
Mushrooms
Olives
Price
Add New

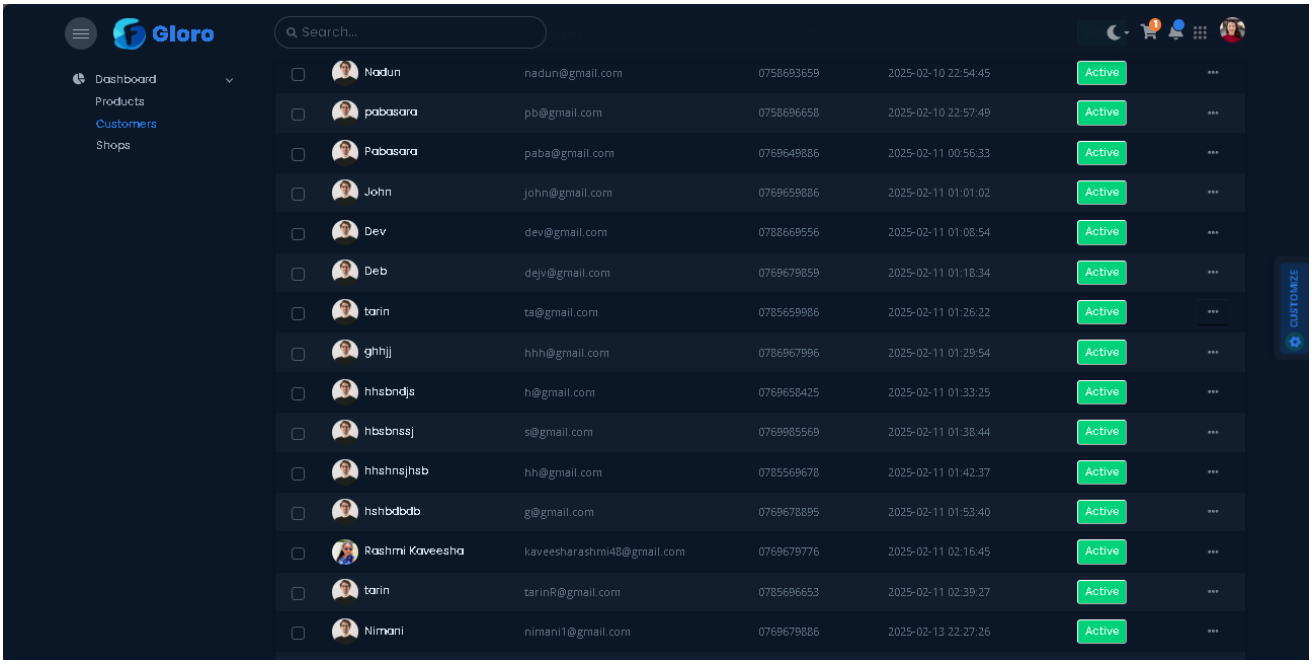

Notification
04 New notifications available
Rashmi Kaveesha
#10049
Arriving
LKR 6650.00
Item :2
Rashmi Kaveesha
#10179
Arriving
LKR 6650.00
Item :2
Rashmi Kaveesha
#14761
Pending
LKR 1100.00
Item :1
Rashmi Kaveesha
#18317
Preparing
LKR 6650.00
Item :2
Rashmi Kaveesha
#19677
Pending
Notifictio...

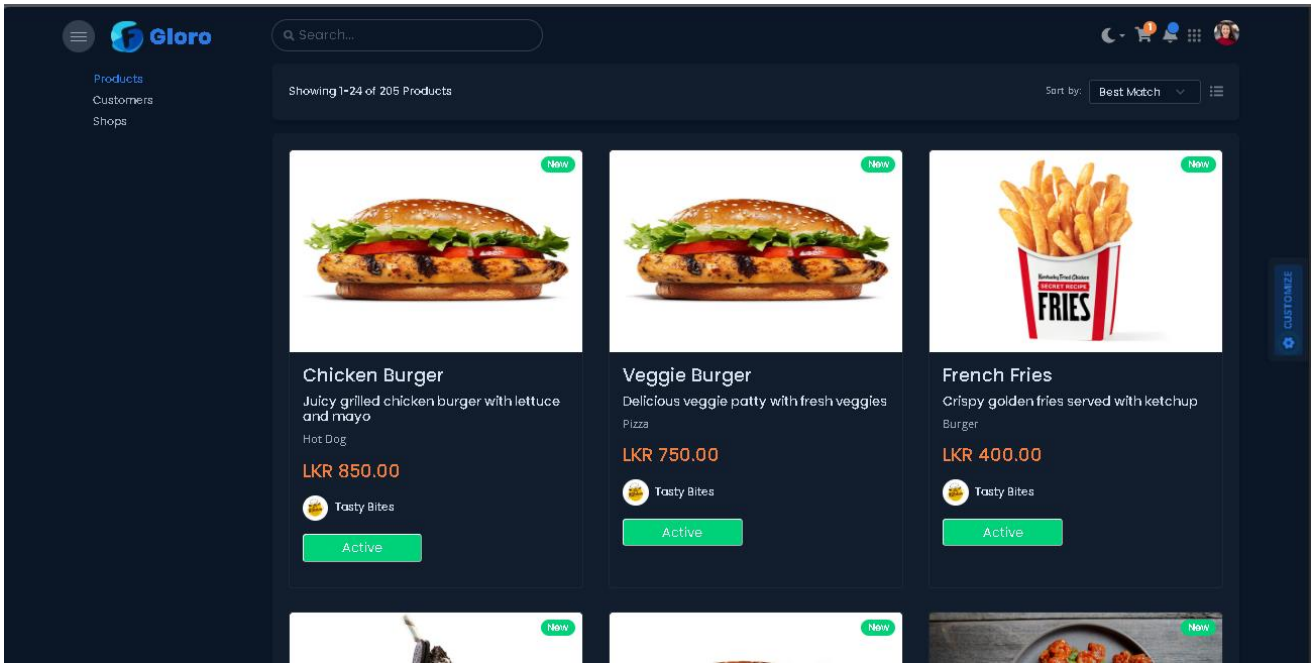

Rashmi Kaveesha : 14761
Rashmi Kaveesha
2025-02-23 13:32:16
LKR 1100.00
Chicken Burger
Id : 76
LKR 850.00
Order Steps
Your order is pending and will be processed soon.
Your order has been reserved.
Your order is being prepared with care.
Your order has been picked up and is on its way.
Your order is arriving soon! Get ready to receive it.
Update to Next Step

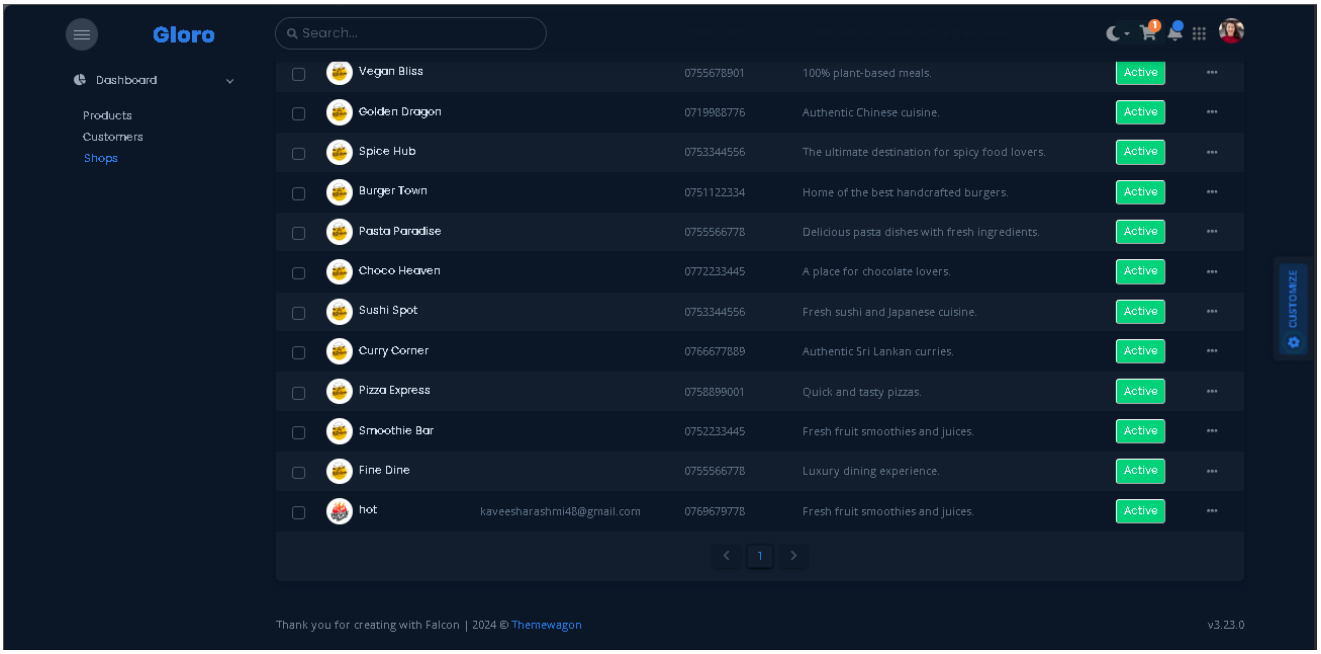

Profile
Manage your all
Available Balance
LKR 50,000.00
Profile Info
Ongoing Orders
Order History
Notification
Customer Reviews
Profile

4.3 Admin Web Dashboard

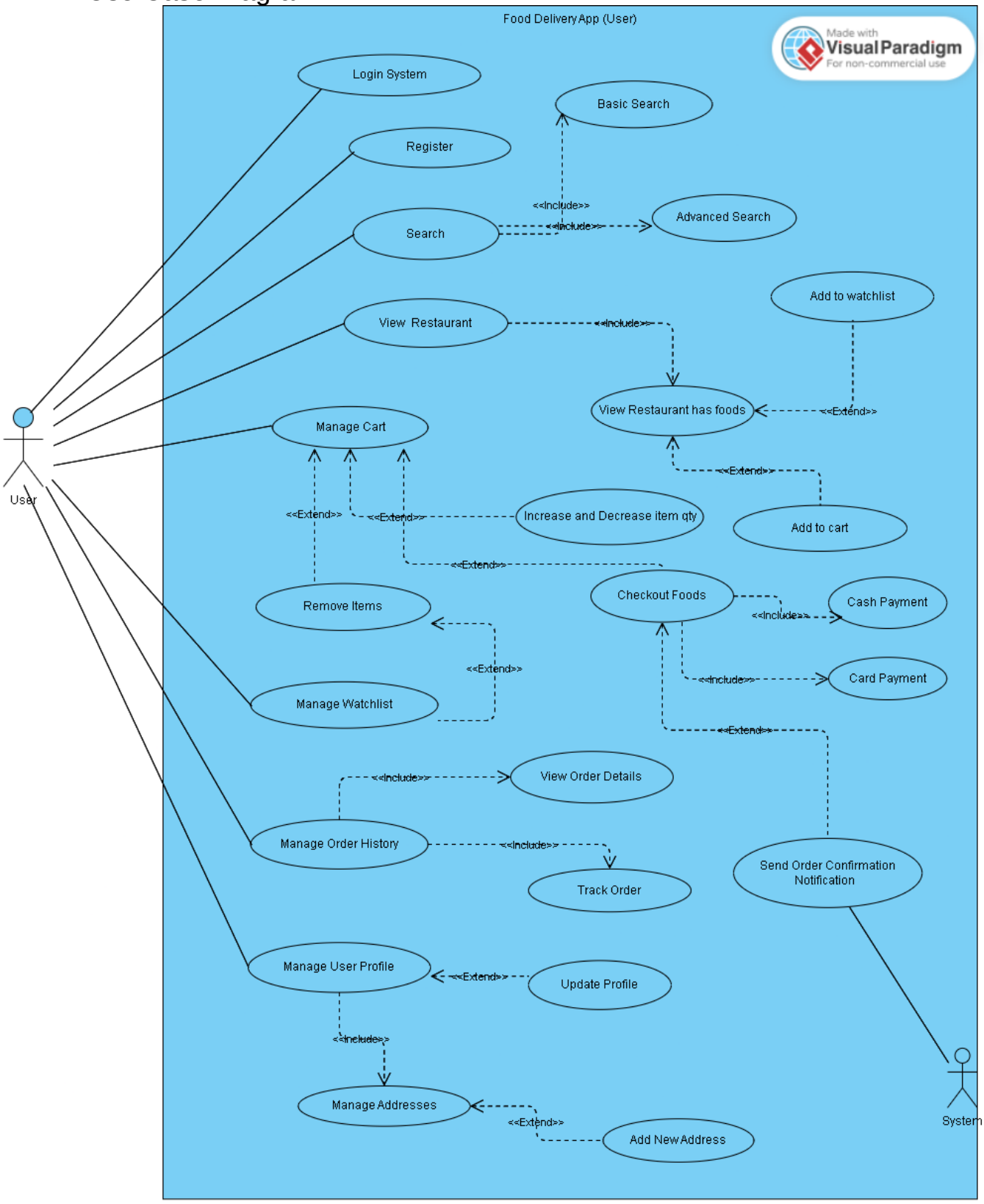


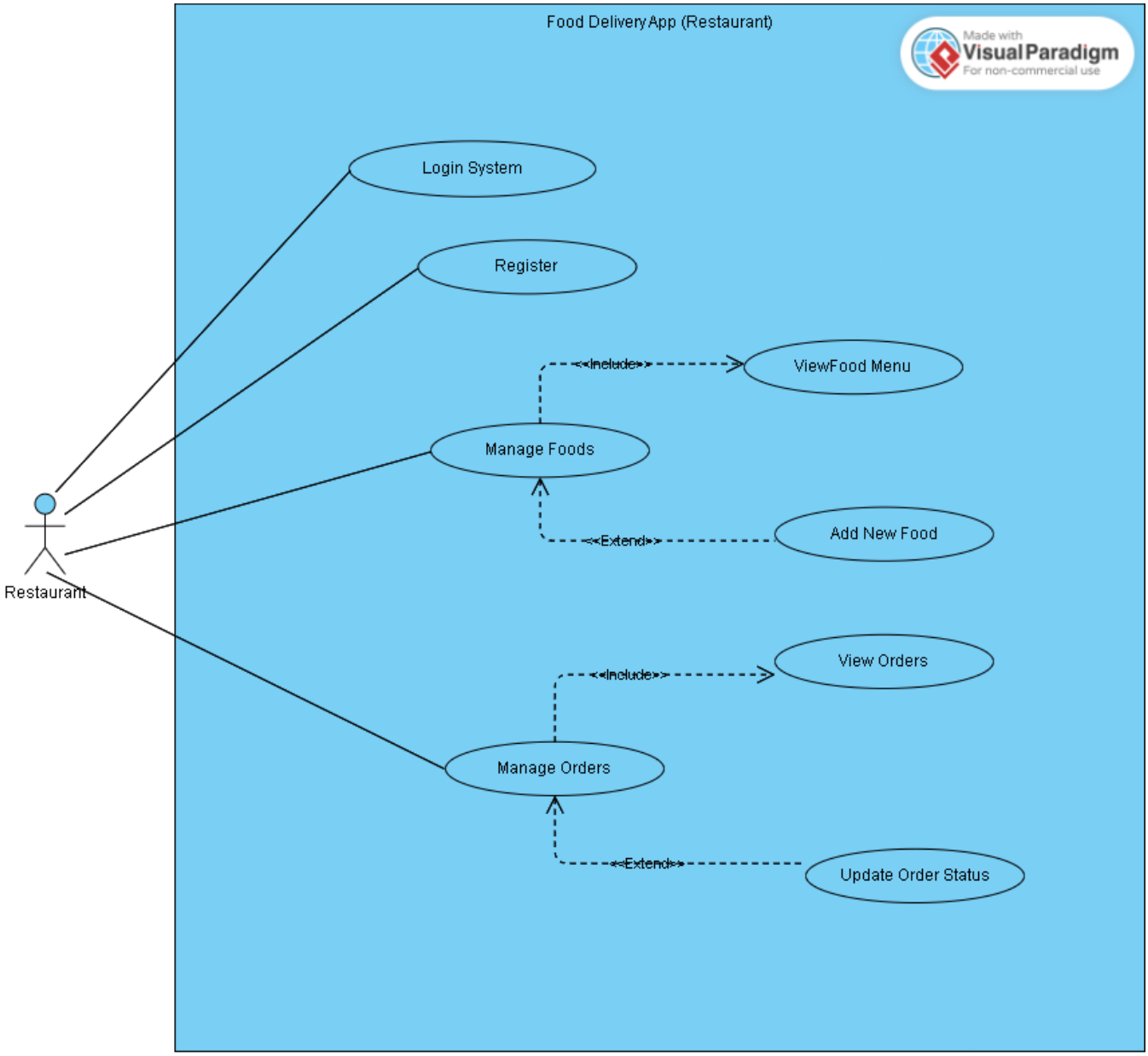


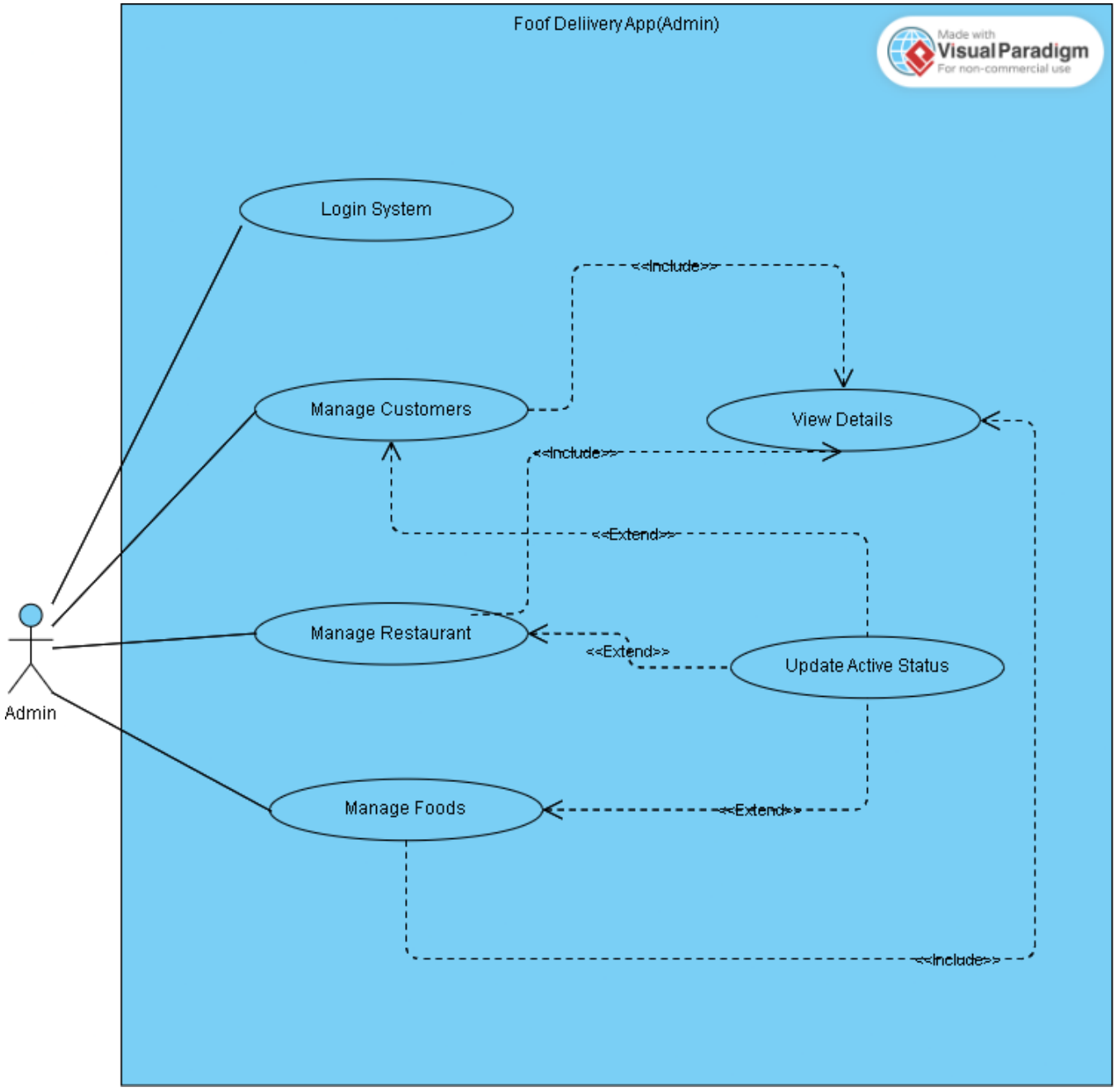

Gloro
Search...
Products
Customers
Shops
Showing 1-24 of 205 Products
Sort by:
Best Match
New
Chicken Burger
Juicy grilled chicken burger with lettuce and mayo
Hot Dog
LKR 850.00
Tasty Bites
Active
Veggie Burger
Delicious veggie patty with fresh veggies
Pizza
LKR 750.00
Tasty Bites
Active
French Fries
Crispy golden fries served with ketchup
Burger
LKR 400.00
Tasty Bites
Active
CUSTOMIZE


Gloro
Search...
Dashboard
Products
Customers
Shops
Vegan Bliss 0755678901 100% plant-based meals. Active
Golden Dragon 0719888776 Authentic Chinese cuisine. Active
Spice Hub 0753344556 The ultimate destination for spicy food lovers. Active
Burger Town 0751122334 Home of the best handcrafted burgers. Active
Pasta Paradise 0755566778 Delicious pasta dishes with fresh ingredients. Active
Choco Heaven 0772233445 A place for chocolate lovers. Active
Sushi Spot 0753344556 Fresh sushi and Japanese cuisine. Active
Curry Corner 0766677889 Authentic Sri Lankan curries. Active
Pizza Express 0758899001 Quick and tasty pizzas. Active
Smoothie Bar 0752233445 Fresh fruit smoothies and juices. Active
Fine Dine 0755566778 Luxury dining experience. Active
hot kaveesharashmi48@gmail.com 0769679778 Fresh fruit smoothies and juices. Active
1
Thank you for creating with Falcon | 2024 © Themewagon
v3.23.0
CUSTOMIZE

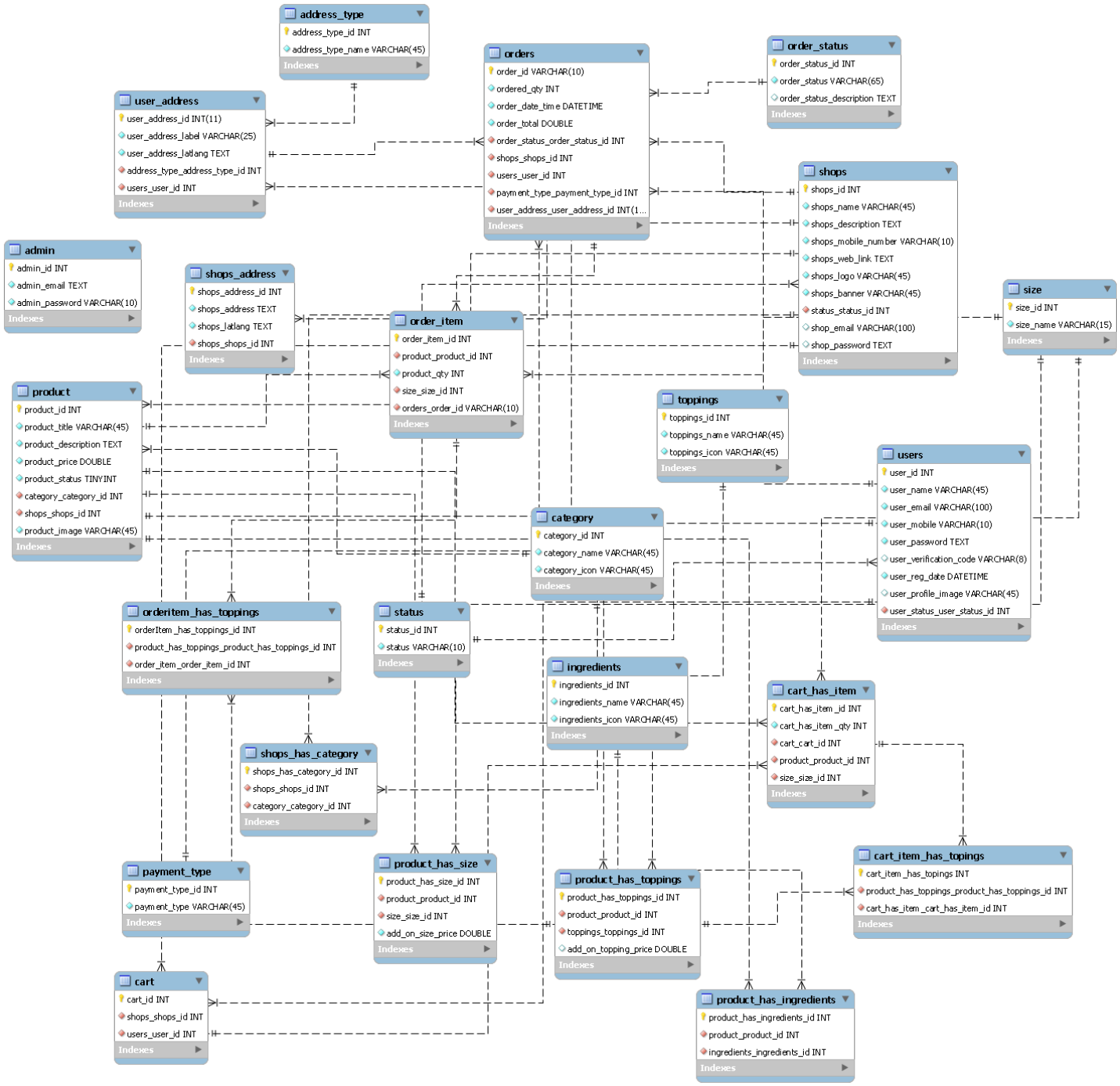
4.4Use Case Diagram




Food Delivery App (Restaurant)
Made with VisualParadigm For non-commercial use
Login System
Register
<<Include>>
ViewFood Menu
Manage Foods
<<Extend>>
Add New Food
Restaurant
View Orders
<<Include>>
Manage Orders
<<Extend>>
Update Order Status


Foof Deliivery App(Admin)
Made with Visual Paradigm For non-commercial use
Login System
<<Include>>
Manage Customers
View Details
<<Include>>
<<Extend>>
Manage Restaurant
<<Extend>>
Update Active Status
Admin
Manage Foods
<<Extend>>
<<Include>>

4.5 Database ER Diagram



5. Implementation

The implementation phase of the GLORO Android app involved translating the system design and specifications into a functional application. This section provides a detailed overview of the development process, including the technologies used and how the app was realized. It also highlights challenges encountered during implementation and the solutions adopted to address them.

5.1Technologies and Tools Used

The following technologies and tools were used to implement the GLORO app:

- **Programming Language**: Java
- **Development Environment**: Android Studio
- **Dependency Management**: Gradle
- **Libraries and Frameworks**:
  - Flexbox: For creating flexible and responsive layouts.
  - ExifInterface: For handling image metadata.
  - Google Play Services Location: For accessing device location.
  - Google Maps SDK: For integrating maps and location-based features.
  - Places API: For finding and autocompleting places.
  - OkHttp: For making HTTP requests.
  - Gson: For JSON parsing and serialization.
  - Glide: For image loading and caching.
  - PayHere SDK: For integrating payment functionality.
  - Firebase Firestore: For cloud-based data storage and synchronization.

5.2 System Architecture

The GLORO app follows the Model-View-ViewModel (MVVM) architecture to ensure separation of concerns and improve maintainability. The key components of the architecture are:

- Model: Represents the data layer, including local database (MySQL) and remote data sources (Firebase Firestore and APIs).
- View: The UI layer, consisting of Activities, Fragments, and XML layouts.
- ViewModel: Acts as a bridge between the View and Model, handling business logic and data manipulation.

6.Results and Evaluation

The GLORO Android app was developed to provide users with a seamless and feature-rich experience for browsing products, making payments, and accessing location-based services. This section evaluates the extent to which the app achieved its goals, summarizes the results of critical tests, and assesses the overall performance and reliability of the system.

6.1 Achievement of Goals

- **User-Friendly Interface:** Fully implemented, providing a clean and intuitive UI with responsive layouts using **Flexbox** and efficient image loading with **Glide**.
- **Product Browsing:** Successfully categorized and displayed products in a **RecyclerView**, with pagination to handle large datasets.
- **Secure Payments:** Integrated **PayHere SDK** for secure and seamless payment processing.
- **Location-Based Services:** Utilized **Google Maps SDK** and **Places API** to provide accurate location tracking and place search functionality.
- **Network Request Handling:** Used **OkHttp** and **Gson** to efficiently handle network requests and JSON parsing.

6.2 Confidence in the System

Based on the results of functional and integration testing, the following conclusions can be drawn about the system's reliability and performance:

- **Core Functionality:** All core features, such as product browsing, payment processing, and location-based services, work as intended and are highly reliable for everyday use.
- **Performance:** The app performs well under normal conditions, with efficient handling of large datasets and network requests.
- **User Experience:** The app provides a smooth and intuitive user experience, with responsive layouts, fast image loading, and user-friendly error handling.
- **Security:** The integration of **PayHere SDK** ensures secure payment processing, and the app follows best practices for data security.

7.Future Work

While the GLORO Android app has successfully achieved its core objectives, there are several additional features and improvements that could be implemented to further enhance the app's functionality and user experience. These ideas represent the next steps for the project and provide a starting point for future development.

7.1 Real-Time Order Tracking

- **Description:** Implement real-time order tracking using **Google Maps API** to allow users to track their orders from the moment they are placed until delivery.
- **Benefits:** Enhances user transparency and satisfaction by providing live updates on the order status and delivery progress.

7.2 Notification System for Order Updates

- **Description:** Send push notifications to users at every step of the order process, such as when the order is confirmed, prepared, out for delivery, and delivered.
- **Benefits:** Keeps users informed about their order status in real-time, improving the overall user experience.

7.3 Multiple Shops Ordering Option

- **Description:** Allow users to place orders from multiple shops in a single transaction, with separate carts for each shop.
- **Benefits:** Provides users with more flexibility and convenience, especially when ordering from different types of shops (e.g., a restaurant and a grocery store).

7.4 Dine-In, Pickup, and Table Booking Options

- **Description:** Add options for users to choose between dine-in, pickup, and delivery. Include a table booking feature for dine-in orders.
- **Benefits:** Expands the app's functionality to cater to different user preferences and scenarios.

7.5 Save Payment Card Options

- **Description:** Implement a feature that allows users to save their payment card details securely for faster checkout in future orders.
- **Benefits:** Improves the user experience by reducing the time required for payment processing.

7.6 Seller Update Products Option

- **Description:** Provide sellers with a dedicated interface to add, update, or remove products from their shop's catalog.
- **Benefits:** Ensures that product information is always up-to-date and reduces the administrative burden on the app's support team.

7.7 Update Shop Profiles

- **Description:** Allow sellers to update their shop profiles, including details such as shop name, description, logo, and contact information.
- **Benefits:** Helps sellers maintain accurate and appealing profiles, which can attract more customers.

7.8 Admin Dashboard for Summary View

- **Description:** Develop an admin dashboard that provides a comprehensive summary of all activities, including orders, payments, user accounts, and shop performance.
- **Benefits:** Enables administrators to monitor and manage the app more effectively, with insights into key metrics and trends.

7.9 Additional Features

- **Loyalty Programs:** Introduce loyalty programs or reward points for frequent users to encourage repeat orders.
- **User Reviews and Ratings:** Allow users to leave reviews and ratings for shops and products, helping others make informed decisions.
- **Multi-Language Support:** Add support for multiple languages to make the app accessible to a broader audience.

8.Conclusions

The GLORO Android app was developed with the aim of creating a robust and user-friendly platform for browsing products, making secure payments, and accessing location-based services. By leveraging a combination of backend technologies, including Java Hibernate, alongside modern mobile development tools, the app successfully achieved its core objectives. This section summarizes the project's aims, main results, and key learnings.

The app's backend was built using Java Hibernate, which provided a reliable and efficient way to manage data persistence and interact with the MySQL database. For real-time data synchronization and cloud-based storage, Firestore was integrated, ensuring that the app could handle dynamic data updates seamlessly. On the device side, SQLite and SharedPreferences were used for local data storage, enabling offline functionality and a smoother user experience. These technologies worked together to create a scalable and efficient data management system.

To enhance the user experience, the app incorporated loading animations to provide visual feedback during data fetching and processing. This ensured that users were kept informed during operations that required waiting, such as loading product lists or processing payments. Additionally, the app included a network availability check to detect whether the device was connected to the internet. If no network was available, the app displayed an alert dialog to inform the user and prompt them to check their connection, ensuring a seamless experience even in offline scenarios.

The development process highlighted the importance of modular design and rigorous testing. By using the MVVM architecture, the app achieved a clear separation of concerns, making it easier to maintain and scale. Testing played a critical role in identifying and resolving issues early, ensuring that the app was reliable and performed well under various conditions. The integration of third-party libraries, such as PayHere SDK for payments and Google Maps SDK for location-based services, further streamlined development and added powerful functionality.

While the app successfully met its core objectives, there are opportunities for future enhancements, such as real-time order tracking, a notification system for order updates, and an admin dashboard for summary views. These features would further improve the app's functionality and user experience. Overall, the GLORO app represents a successful implementation of a modern, user-friendly platform, demonstrating the value of careful planning, modular design, and the effective use of both backend and mobile technologies.

9. Reflection on Learning

Developing the GLORO Android app was not just about mastering technical skills like using Java Hibernate, Firestore, or SQLite, but also about challenging and refining my assumptions and decision-making processes. Initially, I assumed that a robust backend and efficient data handling would be sufficient for a seamless user experience. However, I quickly realized the importance of user-centric design and real-time feedback, such as loading animations and network availability checks, in enhancing usability. This shifted my focus from purely technical solutions to a more holistic approach that balanced functionality with user experience.

Another key learning was the value of modularity and scalability. By adopting the MVVM architecture, I understood how separating concerns could simplify maintenance and future enhancements. This experience reshaped my approach to problem-solving, emphasizing flexibility and long-term planning over quick fixes. Overall, this project not only improved my technical expertise but also deepened my understanding of how assumptions and concepts evolve through practical application, leading to more informed and effective decision-making.

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